Stage-Aware Disambiguation for Subjective Visual Intention Recognition

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Abstract

Visual intention understanding aims to infer the latent purpose conveyed by an image, and is fundamentally more challenging than conventional image classification due to its abstract and subjective nature. A central difficulty of this task lies in two distinct forms of ambiguity: supervisory ambiguity, caused by subjective annotation over plausible intentions, and semantic ambiguity, caused by semantically adjacent and overlapping intents. In this work, we propose SADIR (**Stage-Aware Disambiguation for Intent Recognition**), a framework that treats ambiguity differently across stages: training focuses on stabilizing supervision under disagreement, while inference focuses on discriminating semantically adjacent candidates. To address supervisory ambiguity, we introduce Uncertainty-guided Text Distillation (UTD), which converts annotator disagreement into an adaptive semantic regularization signal. To address semantic ambiguity, we further propose Semantic Local Reranking and Calibration (SLR-C), which performs lightweight reranking within a compact Top-K candidate set using heterogeneous textual priors. Extensive experiments on the Intentonomy dataset demonstrate that SADIR consistently improves both hard-sample robustness and overall multi-label prediction performance over strong baselines and recent state-of-the-art methods.

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1. Introduction

With the rapid proliferation of social media, images are increasingly used as a medium for users to share daily experiences, emotions, and personal viewpoints. Visual intention understanding aims to infer the latent purpose conveyed by an image, and has become an important problem in human-centered multimedia analysis [1, 2]. Unlike explicit textual statements, such visual expressions often imply intentions in an indirect and abstract manner. Accurately recognizing these intentions is valuable for a variety of downstream applications, including user interest understanding, psychological and behavioral profiling, and broader social media content analysis.

Compared with conventional image classification, visual intention recognition is fundamentally more challenging. In traditional recognition tasks, the goal is to assign images to relatively objective semantic categories, such as objects, scenes, or actions. In contrast, intention recognition requires mapping visual evidence to abstract and subjective intent labels. These labels are rarely determined by local visual cues alone; instead, they often emerge from a holistic interpretation of scene context, human activities, object configurations, social interactions, and commonsense knowledge. Therefore, the main difficulty of this task lies not merely in visual representation learning, but in establishing a reliable bridge between concrete image content and abstract intention semantics.

A central challenge of this task is its inherent ambiguity, which manifests in two distinct forms, as shown in Figure 1 (a). The first is semantic ambiguity. Intent labels are often semantically adjacent, partially overlapping, or hierarchically correlated. Different labels may correspond to visually similar patterns, while the same intent may be expressed through diverse appearances. This leads to high inter-class similarity and substantial intra-class variation, making fine-grained discrimination particularly difficult. The second is supervisory ambiguity. Since intention recognition is intrinsically subjective, different

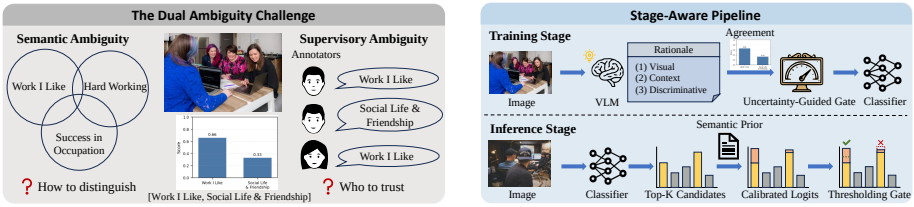

Figure 1: **The Stage-Aware Disambiguation paradigm of SADIR. (a) The Dual Ambiguity Challenge:** Subjective visual intent recognition is fundamentally bottlenecked by *semantic ambiguity* (highly overlapping conceptual boundaries during inference) and *supervisory ambiguity* (annotator disagreement and noise during training). **(b) Stage-Aware Pipeline:** our SADIR framework introduces a *stage-aware* paradigm. During training, the model resolves supervisory ambiguity by leveraging VLM-derived commonsense, adaptively gated by annotator agreement. During inference, the model tackles semantic ambiguity by refining predictions within a compact Top-K candidate pool using text priors, with zero extra visual computational overhead.

audiences may infer different yet plausible intentions from the same image. As a result, the supervisory signal itself may become unstable, especially for difficult samples with severe disagreement. Under such circumstances, directly optimizing against hard labels can force the model to fit oversimplified or unreliable targets, thereby harming the robustness of the learned decision boundaries.

Existing studies have attempted to alleviate these difficulties from different perspectives. Prototype-based methods, such as PIP-Net [3], introduce representative anchors to reduce feature-space confusion. Structure-aware methods, including CPAD [4] and HLEG [5], exploit hierarchical label relationships to enhance multi-level semantic modeling and strengthen supervision. From the perspective of label quality, LabCR [6] explicitly models label shifting and label blemish to better cope with imperfect annotation distributions. More recently, knowledge-enhanced approaches have recognized that visual evidence alone is often insufficient for subjective intention understanding. IntCLIP [7] incorporates the vision-language alignment knowledge of CLIP into the task, while IntentMLM [8] further leverages multimodal large models, caption generation, and retrieval to perform more holistic intention perception.

Although these methods have yielded encouraging progress, they typically treat one perspective of ambiguity as dominant. In our view, the core difficulty of subjective visual intention recognition is not any single ambiguity alone, but the coexistence of supervisory ambiguity and semantic ambiguity, which arise at different stages and affect prediction in different ways. The former primarily corrupts training by compressing diverse subjective interpretations into rigid supervisory targets, whereas the latter primarily affects inference by making semantically adjacent candidate labels difficult to distinguish. Treating these two failure modes within one mechanism may therefore be suboptimal, as it is difficult to simultaneously achieve robust learning on disagreement-prone samples and precise discrimination among closely related intents.

Motivated by this observation, we propose **SADIR** (**S**tage-**A**ware **D**is-ambiguation for **I**ntent **R**ecognition) to address ambiguity in visual intention recognition, as shown in Figure 1 (b). Instead of treating ambiguity as a unified problem, we consider its different roles during training and inference, and design complementary mechanisms accordingly. During training, the primary challenge lies in handling unreliable supervision caused by annotator disagreement. To address this, we propose Uncertainty-guided Text Distillation (UTD), which leverages disagreement as an uncertainty signal to adaptively introduce semantic guidance. This enables the model to learn more robust decision boundaries without overfitting to noisy or oversimplified labels. During inference, the challenge shifts to distinguishing among semantically similar candidate intents. To improve fine-grained discrimination, we introduce Semantic Local Reranking and Calibration (SLR-C), which performs lightweight semantic refinement within a compact candidate set using multiple forms of textual priors. Together, SADIR improves both robustness and discriminability without introducing significant additional complexity.

The main contributions of this work are summarized as follows:

- We analyze visual intention recognition from a stage-dependent perspective and highlight that ambiguity plays different roles during training and

inference. This perspective provides a clearer understanding of why existing methods struggle with both robustness and fine-grained discrimination.

- We propose Uncertainty-guided Text Distillation (UTD), a training strategy that leverages annotator disagreement as an uncertainty signal to adaptively introduce semantic guidance, improving robustness under ambiguous supervision.
- We propose Semantic Local Reranking and Calibration (SLR-C), an inference-time refinement module that enhances discrimination among semantically similar intents through lightweight local reranking with textual priors.

Extensive experiments on the Intentonomy dataset show that SADIR consistently outperforms strong existing baselines and recent state-of-the-art methods, yielding improvements in both hard-sample robustness and overall multi-label prediction quality.

2. Related Work

2.1. Image Intent Recognition

Intent recognition seeks to decode the underlying purposes of humans. Early studies mainly focus on specific domains such as political persuasion [9, 10, 11], where intent is inferred from facial expressions, body language, and scene context. More recently, the release of the Intentonomy dataset [1] has facilitated the study of general-purpose visual intention understanding in unconstrained social media scenarios.

Existing methods for visual intention recognition seek to solve the problem from different perspectives. Methods such as PIP-Net [3] concentrate on feature and representation learning, which introduce prototype-based representations to capture to capture diverse visual patterns and improve interpretability. Considering the hierarchical nature of intent labels, approaches like CPAD [4] and HLEG [5] exploit label hierarchies to improve multi-level semantic alignment

and mitigate ambiguity. LabCR [6] addresses label noise and shifting through calibration strategies, while recent works such as IntCLIP [7] and MLMIntent [8] leverage vision-language models and textual priors to align visual features with semantic intent descriptions.

Despite these advances, most existing methods address ambiguity from a single perspective, such as feature representation, label structure, or external knowledge. They implicitly assume that ambiguity should be handled uniformly within a model. However, in visual intention recognition, ambiguity arises from different sources and may affect different stages of the recognition process in distinct ways.

2.2. Learning under Ambiguous or Uncertain Supervision

Learning under ambiguous or noisy supervision has been extensively studied in multi-label recognition and subjective understanding tasks, where label reliability is often compromised by annotation noise, semantic overlap, or annotator disagreement. Existing approaches typically address this issue through three main directions.

First, noise-robust learning methods mitigate unreliable supervision through robust objectives, hard-example handling, or noise-aware optimization [12, 13, 14, 15]. These methods reduce the adverse impact of corrupted or hard labels, but they usually treat uncertainty as a generic optimization issue rather than modeling its semantic origin.

Second, hierarchical or structure-aware methods improve learning under ambiguous labels by exploiting label relations or taxonomy constraints [16, 5]. Such methods are especially relevant to intention recognition, where semantically nearby labels often exhibit overlapping decision boundaries. However, they mainly focus on label dependency modeling, instead of explicitly using annotator disagreement to regulate supervision strength.

Third, knowledge distillation and vision-language supervision have been adopted to inject additional semantic guidance from teacher models or pretrained language priors [17, 18]. Knowledge distillation transfers structured

knowledge from stronger teacher models, while vision-language models leverage textual descriptions as semantic anchors to enhance visual representations. However, these approaches typically assume that the provided semantic guidance is uniformly reliable across samples, which may not hold in subjective tasks with inherent ambiguity.

In visual intention understanding, ambiguity is intrinsic: the same image may support multiple plausible interpretations, and category boundaries are often fuzzy [1, 5]. Prior work has recognized the importance of hierarchical relations and label ambiguity in this task, but existing methods still largely rely on fixed supervision or globally applied semantic guidance. By contrast, our approach explicitly leverages annotator disagreement as an uncertainty signal, and adaptively regulates semantic guidance according to supervision ambiguity. This results in a more robust learning process under subjective and ambiguous supervision.

3. Method

3.1. Overview

To address the dual challenges of supervisory ambiguity and semantic ambiguity in visual intention recognition, we propose **SADIR** (**S**tage-**A**ware **D**isambiguation for **I**ntent **R**ecognition), which explicitly decouples training and inference. Instead of handling these two forms of ambiguity within a unified mechanism, we address them in a stage-specific manner.

SADIR consists of two key components, as shown in Figure 2:

- **Uncertainty-guided Text Distillation (UTD)**, which introduces adaptive semantic guidance during training to improve robustness under ambiguous supervision;
- **Semantic Local Reranking and Calibration (SLR-C)**, which performs lightweight semantic refinement during inference to enhance discrimination among competing intents.

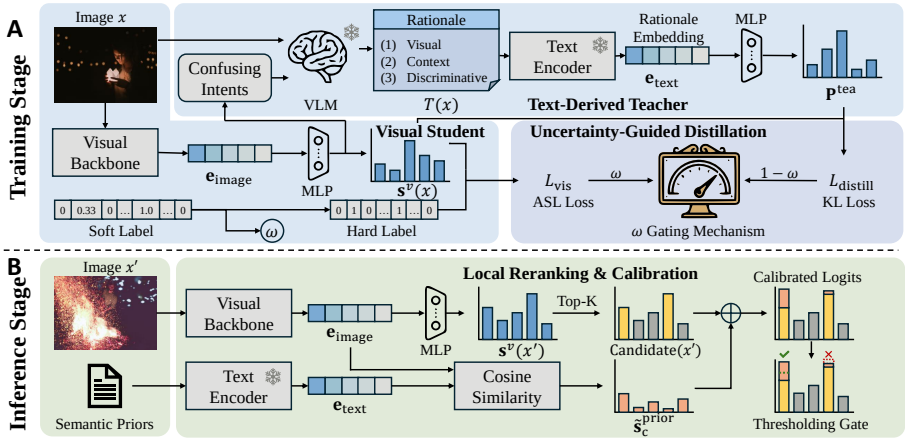

Figure 2: Overview of the SADIR framework. The framework explicitly decouples the disambiguation process into two stages. **(Top) Training Stage:** The Uncertainty-guided Text Distillation (UTD) module employs a VLM-derived teacher to generate contrastive textual rationales. The distillation process is adaptively gated by the annotator agreement score ω, balancing objective visual supervision and subjective semantic guidance. **(Bottom) Inference Stage:** The Semantic Local Reranking and Calibration (SLR-C) module extracts Top-K candidate intents from the visual logits and refines them using visually-grounded semantic priors, followed by class-wise calibration for the final prediction.

Given an input image, the model first produces visual intent scores using a strong visual backbone. UTD is applied during training to shape the representation under uncertain supervision, while SLR-C refines predictions at inference time.

3.2. Problem Formulation

Given an input image x, the goal is to predict a set of intent labels from a predefined label space $\mathcal{Y} = \{1, \ldots, C\}$. Due to the subjective nature of visual intention recognition, annotations are provided as soft label vectors $\mathbf{y}^{\text{soft}} \in \{0, 0.33, 0.66, 1\}^C$, reflecting annotator agreement.

Existing approaches typically follow either hard binarization or soft regression paradigms, both of which have limitations under ambiguous supervision. To address this, we adopt a **decoupled utilization** of annotation signals.

Specifically, we derive binarized labels $\mathbf{y}^{\text{hard}}$ for classification supervision, while extracting an image-level agreement score $\omega \in (0, 1]$ from $\mathbf{y}^{\text{soft}}$. Here, ω reflects annotator consensus, where higher values indicate more reliable supervision.

In practice, an image may contain multiple positive intents with different agreement levels. We therefore adopt a conservative pooling rule and define the sample-level agreement as the minimum soft agreement among all positive labels:

$$\omega = \min_{c:\, y_c^{\text{hard}}=1} y_c^{\text{soft}}. \tag{1}$$

This choice lets the hardest positive label determine how much the sample should trust direct supervision.

This agreement score is used to adaptively control the contribution of semantic guidance during training. The model outputs a logit vector $\mathbf{s}^v \in \mathbb{R}^C$, which is later refined and thresholded for final prediction.

3.3. Uncertainty-guided Text Distillation (UTD)

3.3.1. Motivation

Supervisory ambiguity arises when multiple plausible intents coexist for a given image. Direct optimization against hard labels may lead to overconfident predictions and unstable decision boundaries.

To alleviate this issue, we introduce semantic guidance from text space and adapt its influence based on the uncertainty of supervision.

3.3.2. Text-derived Teacher

For each training image, we construct structured textual descriptions that capture semantic cues relevant to intention understanding. These descriptions are generated using a vision-language model, guided by both the image content and intent semantics.

To enhance discriminative capability under semantic ambiguity, we further incorporate contrastive cues by identifying visually plausible but incorrect intents predicted by the current model. Specifically, we select high-confidence

negative intents as confusion candidates and use them to guide the generation of discriminative textual descriptions. This encourages the resulting text to emphasize subtle distinctions among semantically adjacent intents.

The generated text aims to summarize three aspects: (1) observable visual elements, (2) contextual interpretation beyond direct perception, and (3) discriminative cues for distinguishing confusing intents.

The resulting text $T(x)$ is encoded into a dense embedding space using a frozen text encoder:

$$\mathbf{e}_{\text{text}} = f_{\text{text}}(T(x)). \tag{2}$$

A lightweight classifier then maps the embedding into a multi-label teacher distribution:

$$\mathbf{p}^{\text{tea}} = \sigma(g_{\text{tea}}(\mathbf{e}_{\text{text}})/\tau), \tag{3}$$

where $\sigma(\cdot)$ is the sigmoid function and τ is the temperature parameter.

This design introduces contrastive semantic signals that improve the teacher's ability to distinguish semantically similar intents, while remaining independent of the inference procedure.

3.3.3. Uncertainty-guided Distillation

The visual backbone produces logits $\mathbf{s}^v$ for image x. We adopt Asymmetric Loss [15] for classification:

$$\mathcal{L}_{\text{vis}} = \frac{1}{C}\sum_{c=1}^{C} \ell_{\text{ASL}}(s_c^v, y_c^{\text{hard}}). \tag{4}$$

To incorporate semantic guidance, we introduce an uncertainty-guided Bernoulli distillation loss:

$$\mathcal{L}_{\text{distill}} = \sum_{c=1}^{C} \text{KL}\big(\sigma(s_c^v/\tau) \,\|\, p_c^{\text{tea}}\big), \tag{5}$$

which is applied independently to each label. We use Bernoulli-style distillation rather than softmax distillation because intent recognition is a multi-label problem and class probabilities are not mutually exclusive. Here, λ denotes the distillation weight that controls the overall contribution of the teacher signal relative to the visual supervision term. In our experiments, λ is fixed to 1.0.

Rather than mixing the two objectives with a fixed global coefficient, we use a dynamic gate derived from the agreement score. For the current sample, the teacher branch weight is defined as

$$g = 1 - \omega, \tag{6}$$

so low-agreement samples receive stronger semantic guidance, while high-agreement samples remain dominated by direct visual supervision.

Using g, the final loss is

$$\mathcal{L}_{\text{total}} = (1 - g)\mathcal{L}_{\text{vis}} + \lambda g \mathcal{L}_{\text{distill}} = \omega \mathcal{L}_{\text{vis}} + \lambda(1 - \omega)\mathcal{L}_{\text{distill}}. \tag{7}$$

This dynamic-gated formulation allows the model to rely more on visual supervision when annotations are reliable, and to leverage semantic guidance when supervision is uncertain. In other words, the gate is not an inference-time module, but a training-time routing mechanism that adaptively reallocates optimization weight between hard-label supervision and text-teacher distillation.

3.4. Semantic Local Reranking and Calibration (SLR-C)

While UTD improves representation robustness, semantic ambiguity still affects final predictions, especially among visually similar intents. To address this, we propose SLR-C for inference-time refinement.

3.4.1. Semantic Prior Construction

Intent semantics are inherently multi-granular and abstract, making a single textual representation insufficient. We therefore construct a heterogeneous semantic prior for each intent by combining three complementary textual sources: lexical intent phrases, canonical intent descriptions, and descriptive visual scenarios. The lexical source provides concise intent name cues, the canonical source provides semantic definitions, and the scenario source grounds abstract intents into plausible visual situations.

Given an image x, we compute image-text similarities:

$$s_c^{\text{prior}} = \text{sim}(f_{\text{img}}(x), \mathbf{e}_c). \tag{8}$$

In practice, $\mathrm{sim}(\cdot,\cdot)$ is cosine similarity between L2-normalized image and text embeddings, scaled by the CLIP logit scale γ, i.e.

$$\mathrm{sim}(f_{\mathrm{img}}(x), \mathbf{e}_c) = \gamma \cdot \frac{f_{\mathrm{img}}(x)^\top \mathbf{e}_c}{\|f_{\mathrm{img}}(x)\|_2 \|\mathbf{e}_c\|_2}. \tag{9}$$

For each source, we obtain a class-wise similarity score and normalize it independently. Specifically, for source $r \in \{\mathrm{lex}, \mathrm{can}, \mathrm{scen}\}$, we perform per-sample normalization across all classes:

$$\tilde{s}_c^r = \frac{s_c^r - \mu^r(x)}{\sigma^r(x) + \epsilon}, \tag{10}$$

where $\mu^r(x) = \frac{1}{C}\sum_{c=1}^{C} s_c^r$ and $\sigma^r(x)$ is the corresponding standard deviation over the C intent classes, with ϵ a small constant for numerical stability. The final prior is then constructed by averaging the normalized scores from the three sources:

$$\tilde{s}_c^{\mathrm{prior}} = \frac{1}{3}\left(\tilde{s}_c^{\mathrm{lex}} + \tilde{s}_c^{\mathrm{can}} + \tilde{s}_c^{\mathrm{scen}}\right). \tag{11}$$

3.4.2. Semantic Local Reranking

We first select a compact candidate set:

$$\mathcal{C}_K(x) = \mathrm{TopK}(\mathbf{s}^v, K). \tag{12}$$

The logits are refined via local adjustment:

$$\tilde{s}_c = \begin{cases} s_c^v + \alpha \cdot \tilde{s}_c^{\mathrm{prior}}, & c \in \mathcal{C}_K(x), \\ s_c^v, & \text{otherwise}. \end{cases} \tag{13}$$

This localized refinement focuses on resolving ambiguity among competing intents without disturbing the global prediction structure.

3.4.3. Class-wise Calibration

To account for class imbalance and varying score distributions, we adopt class-wise thresholds:

$$\hat{y}_c = \mathbb{I}[\sigma(\tilde{s}_c) > t_c], \tag{14}$$

where t_c is determined on the validation set. Concretely, each threshold is selected independently by maximizing the class-wise validation F1:

$$t_c^* = \arg\max_{t \in \mathcal{T}} \mathrm{F1}_c^{\mathrm{val}}(t), \tag{15}$$

where $\mathcal{T}$ is a fixed threshold grid. This design lets each class adapt its decision boundary to its own score distribution and frequency level, instead of sharing a single global threshold.

3.5. Stage-Aware Ambiguity Modeling

We interpret visual intention recognition as a stage-dependent ambiguity problem. Supervisory ambiguity primarily affects representation learning, while semantic ambiguity affects decision boundaries.

SADIR addresses these two aspects in a stage-aware manner: UTD improves robustness during training through uncertainty-guided semantic distillation, while SLR-C enhances discrimination during inference via localized semantic refinement.

This stage-aware design enables effective handling of ambiguity without introducing heavy architectural complexity.

4. Experiments

4.1. Experimental Setup

4.1.1. Dataset and Evaluation Metrics

We evaluate our method on the Intentonomy dataset [1], a multi-label benchmark for visual intention recognition in social media scenarios. Following the standard split used in our experiments, the dataset contains 12,739 training images, 498 validation images, and 1,216 test images. Intentonomy additionally provides soft agreement annotations with values in $\{0, 0.33, 0.66, 1\}$ on the training set, which directly reflect the degree of annotator consensus and serve as the uncertainty signal ω in our framework.

Table 1: Comparison with state-of-the-art methods on the Intentonomy test split. We report Macro F1, Micro F1, Samples F1, and Avg. (the mean of the three F1 scores). All numbers are percentages.

Method	Publication	Macro F1	Micro F1	Samples F1	Avg.
Random	-	6.94	7.18	7.10	7.07
Visual [19]	CVPR 2016	22.77	30.23	28.45	27.15
Multi-label classification methods					
HiMulConE [16]	CVPR 2022	28.43	41.50	42.15	37.36
MultiGuard [20]	NeurIPS 2022	26.87	38.67	40.06	35.20
SST [21]	AAAI 2022	27.33	42.21	42.62	37.39
Query2Label [22]	-	32.12	44.64	45.15	40.64
DualCoOp [23]	NeurIPS 2022	31.06	44.42	40.40	38.63
Label noise methods					
Cleannet [24]	CVPR 2018	25.89	42.15	40.97	36.34
LS [25]	ICML 2020	24.32	42.77	43.27	36.79
Dividemix [26]	ICLR 2020	30.11	40.23	41.56	37.30
Intent perception methods					
Jia [1]	CVPR 2021	23.98	31.28	31.39	28.88
CPAD [4]	TIP 2023	27.39	40.98	41.12	36.50
HLEG [5]	TAC 2023	32.77	44.69	45.54	41.00
PIP-Net [3]	TMM 2023	32.57	45.94	47.05	41.85
LabCR [6]	TIP 2024	34.63	48.51	48.05	43.73
IntCLIP [7]	ECCV 2024	40.15	53.54	51.01	48.23
IntentMLM [8]	PR	43.05	54.77	56.75	51.52
CLIP ViT-L/14 [18]	-	47.40	56.42	56.41	53.41
SADIR (UTD only)	-	50.31	57.97	59.07	55.78
SADIR	-	**51.21**	**59.64**	**60.43**	**57.09**

We report Macro F1, Micro F1, Samples F1. For consistency with recent Intentonomy papers, the main comparison table additionally reports *Avg.* which is the average of three F1 scores.

To better understand the effectiveness of our method, we further conduct

experiments on subsets of the intent classes. We follow the work of Jia et al. [1] and break down intent classes according to content difficulty and dependency. Difficulty classes are measured by how much a standard ResNet-50 model outperforms the random results. Content classes are divided into object-dependent (*Object*-classes), context-dependent (*Context*-classes) and *Others* which depends on both foreground and background information. We emphasize the *Hard* score throughout the analysis because it specifically measures performance on ambiguity-prone intent categories, where semantic overlap and subjective disagreement are most severe. Improvements on the hard subset therefore provide stronger evidence that a method is resolving the core difficulty of intention understanding rather than merely exploiting easy classes.

4.1.2. Implementation Details

Our default visual backbone is frozen CLIP ViT-L/14 [18]. The visual student is implemented as a lightweight residual MLP trained on top of a fixed SLR-C prior. For UTD, rationale text is generated offline by Qwen2.5-VL-7B-Instruct [27] and then encoded by a frozen bge-large-en-v1.5 text encoder [28]. The rationale-generation prompt template is provided in Appendix A.1. The teacher consumes only the resulting text embeddings and is optimized with Asymmetric Loss (ASL) [15] using $\gamma_- = 2$, $\gamma_+ = 0$, and clipping value 0.05. The student is optimized with AdamW [29], learning rate 5×10^{-4}, weight decay 10^{-4}, batch size 256, and dropout 0.1. The teacher hidden dimension is 1024 and the residual student hidden dimension is 768. The distillation temperature is set to 2.0. In SLR-C, priors are constructed from three sources, namely lexical intent phrases, canonical intent descriptions, and Gemini 3.1 Pro-generated scenario descriptions [30] with frozen CLIP text encoder [18]; their normalized logits are averaged before local reranking within Top-5 visual candidates. The scenario-prompt template is also provided in Appendix A.2. During inference, class-wise thresholds are searched on the validation set and then fixed for test evaluation. Notably, UTD introduces *zero extra inference cost*, since the text teacher is only used during training and is discarded at test time.

4.2. Comparative Experiments

4.2.1. Comparison with state-of-the-art methods

Table 1 summarizes the main comparison on the testing set. The strong CLIP baseline already surpasses the previously reported best result by a clear margin, confirming that visual representation is no longer the primary bottleneck for this task. Once the backbone is sufficiently strong, the remaining challenge is how to map rich visual semantics into abstract and subjective intent labels. More importantly, *UTD only* already improves over IntentMLM by a large margin while introducing zero extra inference cost, showing that robust training under subjective supervision is by itself sufficient to outperform recent multimodal large-model pipelines. Building on this, the complete SADIR framework with lexical-canonical-scenario priors further improves all four reported metrics on the testing split, reaching the strongest overall average score under this final configuration. This pattern is consistent with our motivation: SLR-C improves semantic discrimination at inference, whereas UTD stabilizes training under ambiguous supervision.

4.2.2. Performance on Different Difficulty Subsets

To better understand the effectiveness of our method, we further evaluate performance on intent subsets with different difficulty levels. Following Jia et al. [1], we divide intent classes into *Easy*, *Medium*, and *Hard* groups according to how much a standard ResNet-50 classifier outperforms random prediction. For previously published methods, we directly adopt the reported values from the difficulty-wise comparison figure provided in the original papers. For our models, the numbers are computed from the corresponding evaluation logs.

The difficulty-wise results provide two additional insights. First, our baseline is already substantially stronger than prior methods on all three difficulty groups, indicating that the new performance regime is not limited to a specific subset of classes. Second, SADIR further improves medium and hard groups most consistently, which is precisely where semantic ambiguity and supervision uncertainty are more pronounced.

Table 2: Performance on difficulty-based intent subsets (Easy/Medium/Hard) on the Intentonomy test split. Avg. denotes the mean over the three subsets. All numbers are percentages.

Method	Easy	Medium	Hard	Avg.
Random	19.86	7.11	2.81	9.93
Visual [19]	54.64	24.92	10.71	30.09
Jia (w/o hashtag) [1]	57.10	25.68	12.72	31.83
Jia (hashtag) [1]	58.86	26.30	13.11	32.76
Query2Label [22]	81.59	38.06	14.55	44.73
CPAD [4]	69.47	31.50	8.54	36.50
HLEG [5]	81.49	38.99	20.22	46.90
PIP-Net [3]	70.73	33.63	14.33	39.56
IntentMLM [8]	76.26	46.87	27.39	50.17
CLIP ViT-L/14 [18]	80.99	53.32	28.45	54.25
SADIR	**81.01**	**57.06**	**33.48**	**57.18**

4.2.3. Performance of Different Content-Dependent Subsets

Table 3 reports results on content-dependent subsets. Following the protocol in prior work, intent classes are divided into *Object-dependent*, *Context-dependent*, and *Others*. The first group mainly depends on salient foreground entities, the second relies more strongly on scene or environmental context, and the third requires combined reasoning over both foreground and background cues.

The results again show that the strong CLIP baseline is already highly effective on object- and context-dependent intents. The major gain of SADIR appears in the *Others* subset, where intent recognition requires more holistic reasoning over multiple weak cues rather than direct reliance on a single object or scene pattern.

Table 3: Performance on content-dependent intent subsets (Object, Context, and Others) on the Intentonomy test split. Avg. denotes the mean over the three subsets. All numbers are percentages.

Method	Object	Context	Others	Avg.
Random	7.75	12.53	6.05	8.78
Visual [19]	25.58	30.16	21.34	25.69
Jia (w/o hashtag) [1]	28.15	28.62	22.60	26.46
Jia (hashtag) [1]	29.66	32.48	22.61	28.25
CPAD [4]	28.81	47.30	24.74	33.62
PIP-Net [3]	39.20	39.53	27.59	35.44
IntentMLM [8]	50.13	49.69	39.75	46.52
CLIP ViT-L/14 [18]	61.80	61.69	40.59	54.69
SADIR	**62.73**	**62.22**	**45.80**	**56.92**

4.3. Ablation Studies

4.3.1. The Effectiveness of Rationale

We first compare generic image captioning with rationale-style supervision under the same final distillation protocol. We then further ablate the internal structure of the rationale used by the text teacher. Specifically, we compare Caption, which only describes visible content in short natural language; Step 1 only, which contains observable visual evidence; Step 1 with Step 2, which additionally introduces contextual interpretation; and Full rationale, which further adds counterfactual disambiguation cues. Table 4 reports the results under dynamic gated distillation.

The trend is revealing. Caption supervision is already a reasonably strong privileged signal, but it remains slightly weaker than the full rationale in Macro, Samples, Avg., and Hard. Step 1 already provides useful visual facts, but adding Step 2 alone slightly degrades the overall classification metrics, suggesting that contextual interpretation without explicit disambiguation may introduce semantic drift. Once the counterfactual Step 3 is added, however, the rationale be-

Table 4: Ablation of rationale construction for UTD on the Intentonomy test split. Step 1/2/3 correspond to visual evidence, contextual bridging, and counterfactual disambiguation. All numbers are percentages.

Method	Macro	Micro	Samples	Avg.	Hard
Caption	49.64	**58.64**	58.29	55.52	29.14
Rationale (Step 1)	49.81	56.39	57.65	54.62	29.28
Rationale (Step 1 & 2)	49.69	54.45	55.97	53.37	28.82
Rationale (Step 1 & 2 & 3)	**50.31**	57.97	**59.07**	**55.78**	**30.24**

Table 5: Effect of agreement-aware gated distillation on the Intentonomy test split (UTD without SLR-C). All numbers are percentages.

Method	Macro	Micro	Samples	Avg.	Hard
Baseline	47.40	56.42	56.41	53.41	28.45
UTD (w/o gated)	49.59	**58.26**	58.88	55.58	**30.44**
UTD (with gated)	**50.31**	57.97	**59.07**	**55.78**	30.24

comes significantly more effective. This supports our central claim that subjective intention recognition requires not only richer context, but also explicit boundary contraction among confusing alternatives.

4.3.2. Effectiveness of Gated Distillation

We next isolate the effect of UTD without using the SLR-C prior. Table 5 compares plain supervised learning, distillation without gating, and agreement-aware gated distillation.

The results show that both distillation variants improve substantially over plain supervised learning, but agreement-aware gating provides the best overall balance in Macro and Samples F1. This supports our interpretation that dynamic gating acts as a form of *on-demand distillation*: it suppresses over-regularization on reliable samples while reserving teacher guidance for genuinely ambiguous cases. The complementary evidence on weaker backbones is reported later in the robustness analysis.

Table 6: Teacher modality ablation on the Intentonomy test split. Oracle denotes teacher-only performance, while Dynamic KD reports student performance with gated distillation. All numbers are percentages.

Method	Image	Text	Macro	Micro	Samples	Avg.	Hard
Oracle teacher		✓	45.80	51.42	52.61	49.94	**30.21**
Oracle teacher	✓	✓	45.87	53.44	54.04	51.12	28.08
Gated student		✓	**50.31**	**57.97**	**59.07**	**55.78**	**30.24**
Gated student	✓	✓	48.57	56.10	57.13	53.93	28.58

4.3.3. Modality Collapse Avoidance

We further study whether the teacher should consume both image and text features, or only text features. Table 6 reports teacher-side oracle performance and student-side dynamic distillation performance as two separate method groups.

Interestingly, the multimodal oracle teacher is not weaker by itself, but it becomes a worse distillation target. This indicates that in a subjective recognition task, explicit image-text fusion inside the teacher may introduce noisy or redundant visual evidence, whereas a text-only semantic manifold provides cleaner dark knowledge for the student.

4.3.4. Effectiveness of SLR-C and Residual Refinement

We next study how SLR-C interacts with residual learning and distillation. Table 7 shows that prior alone is insufficient, while adding a trainable student and distillation is essential for recovering robust intent predictions.

Two observations are noteworthy. First, naively injecting a heterogeneous prior is not enough; without a trainable student, the mixed prior can distort the score geometry and lead to poor predictions. Second, once distillation is introduced, the residual student becomes consistently beneficial, showing that heterogeneous prior fusion and uncertainty-aware supervision must be optimized jointly rather than independently.

Table 7: Component ablation of SADIR on the Intentonomy test split. *w/o UTD* removes uncertainty-guided text distillation, and *w/o Gated* removes agreement-aware gating. All numbers are percentages.

Method	Macro	Micro	Samples	Avg.	Hard
w/o UTD	47.90	55.98	55.18	53.02	29.71
w/o Gated	49.69	57.90	58.98	55.52	31.69
SADIR (full)	**51.21**	**59.64**	**60.43**	**57.09**	**33.48**

Table 8: Ablation of lexical, canonical, and scenario textual priors in SLR-C on the Intentonomy test split. Ensemble averages normalized logits from selected priors. All numbers are percentages.

Lexical	Canonical	Scenario	Macro	Micro	Samples	Avg.	Hard
✓			50.66	57.23	57.34	55.08	33.28
	✓		50.25	57.27	57.22	54.91	33.23
		✓	51.28	**59.13**	58.47	**56.29**	33.98
✓	✓		50.65	57.37	57.46	55.16	32.38
✓	✓	✓	**51.49**	58.76	**58.54**	56.26	**35.08**

4.3.5. Heterogeneous Priors in SLR-C

Finally, we compare different textual priors used by SLR-C. We consider three semantic sources: *Lexical*, which uses short class-name phrases; *Canonical*, which uses the official intent descriptions provided by Intentonomy; and *Scenario*, which uses Gemini-generated visual scenarios. For the ensemble setting, we average the normalized logits from the selected sources before local reranking. Table 8 shows that the three-source ensemble offers the strongest hard-case robustness and is therefore adopted in the final method. This table isolates the effect of prior sources under the SLR-C reranking stage only, without the residual student or UTD.

This result is consistent with our semantic interpretation of the task. Lexical prior is compact but under-expressive, canonical prior is more descriptive but still abstract, while scenario prior translates latent intention into typical visual

Table 9: Sensitivity to candidate size K on the Intentonomy test split. Thresholding is fixed to class-wise search on the validation split. Avg. denotes the mean of Macro, Micro, and Samples F1. All numbers are percentages.

Top-K	Macro	Micro	Samples	Avg.	Hard
5	**51.21**	**59.64**	**60.43**	**57.09**	**33.48**
10	48.78	57.65	58.25	54.89	29.33
28	50.02	59.33	59.77	56.37	31.21

situations and therefore best bridges the gap between visual content and subjective semantics. The lexical-canonical ensemble remains competitive, showing that heterogeneous priors are beneficial. When all three sources are combined, the resulting ensemble further improves hard-case robustness and offers the broadest semantic coverage across prior types, which motivates our final choice of lexical-canonical-scenario priors in SADIR.

4.3.6. Sensitivity to Candidate Size K

We next study the effect of candidate size K while fixing the thresholding strategy to the final class-wise setting. Table 9 reports the test-split results obtained by retraining SADIR under $K \in \{5, 10, 28\}$ and then searching thresholds on the validation split. The comparison shows that the best overall result is achieved at $K = 5$, while larger candidate sets remain competitive but slightly weaker. This suggests that, under the final lexical-canonical-scenario prior, a compact local candidate set is sufficient and can even be preferable to broader reranking ranges.

4.3.7. Class-wise Thresholding

We then isolate the effect of thresholding under the best-performing candidate setting ($K = 5$). Table 10 compares global and class-wise thresholding on the test split, with both thresholds searched on the validation split. The difference is substantial: global thresholding yields a more permissive decision rule and is weaker overall, whereas class-wise thresholding improves all reported

Table 10: Effect of thresholding strategy on the Intentonomy test split with $K = 5$. Thresholds are searched on the validation split. Avg. denotes the mean of Macro, Micro, and Samples F1. All numbers are percentages.

Thresholding	Macro	Micro	Samples	Avg.	Hard
Global	46.24	56.07	56.00	52.77	26.94
Class-wise	**51.21**	**59.64**	**60.43**	**57.09**	**33.48**

Table 11: Backbone sensitivity of SADIR on the Intentonomy test split. All numbers are percentages.

Backbone	Method	Macro	Micro	Samples	Avg.	Hard
ResNet101	Baseline	36.56	41.87	43.10	40.51	18.99
ResNet101	SADIR	41.27	46.50	47.63	45.13	22.37
CLIP ViT-L/14	Baseline	47.40	56.42	56.41	53.41	28.45
CLIP ViT-L/14	SADIR	**51.21**	**59.64**	**60.43**	**57.09**	**33.48**

metrics, including a large gain on the hard subset. This confirms that per-class calibration is a core part of the final SADIR decision rule rather than a cosmetic post-processing step.

4.4. Backbone Independence

To examine whether SADIR depends entirely on a strong backbone, we also test the full pipeline with a weaker ResNet101 student. The results are shown in Table 11. The full method remains functional, but the gains become smaller on the weaker backbone. This suggests that the framework is portable across backbones, yet its full benefit is realized only when the underlying visual representation is already reasonably strong.

4.5. Efficiency Comparison

Finally, we compare the inference cost of the baseline, SADIR (UTD only), and the full SADIR pipeline. Since all three methods share the same frozen CLIP ViT-L/14 feature extractor, Table 12 reports the *additional* inference

Table 12: Efficiency comparison on top of frozen CLIP features. Params count inference-active heads only; FLOPs are reported per image; latency is measured with batch size 1 on a single RTX 4090.

Method	Params (M)	FLOPs (M)	Latency (ms)
Baseline	1.20	2.40	0.109
SADIR (UTD only)	1.20	2.40	0.109
SADIR	2.41	4.93	0.466

cost on top of pre-extracted CLIP features. Parameters count the inference-active prediction heads only, FLOPs are computed per image under the same feature-level protocol, and latency is measured with batch size 1 on a single RTX 4090 GPU.

The table confirms two intended properties of our design. First, SADIR (UTD only) has exactly the same inference cost as the baseline, because the text teacher is used only during training and discarded at test time. Second, the full SADIR pipeline introduces additional reranking and residual refinement, but the absolute overhead remains small: even with the full lexical-canonical-scenario prior and residual head, the feature-level inference latency stays well below 1 ms per image. This shows that the final performance gains are not obtained by paying a large deployment cost.

4.6. Case Study

Figure 3 provides a qualitative comparison between the CLIP baseline and SADIR. Each row summarizes the input image together with its soft label, model predictions and supporting rationale. These examples help clarify both where SADIR gains its robustness and where residual ambiguity remains difficult even for the full model.

Two observations are particularly useful. First, the top row illustrates the main advantage of rationale-guided training: once the teacher explicitly states why a confusing option should be rejected, the student learns a sharper semantic boundary and avoids the baseline's over-prediction. This behavior is con-

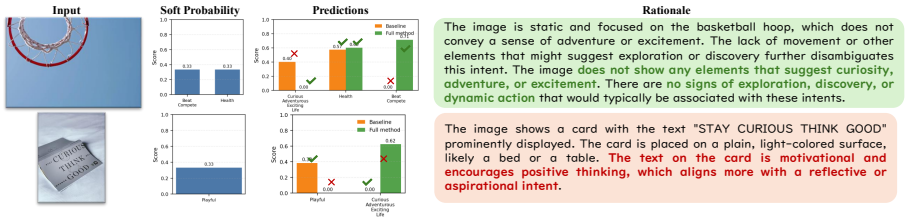

Figure 3: Qualitative case study of baseline and SADIR on representative Intentonomy samples. In the first row, the baseline predicts an incorrect semantic alternative, whereas SADIR removes that distractor by leveraging counterfactual evidence in the rationale and yields the correct intent decision. In the second row, SADIR makes an error, but the annotated soft label itself has only 0.33 agreement, indicating substantial ambiguity; under such weak supervision, the alternative prediction made by SADIR remains semantically plausible rather than arbitrarily wrong.

sistent with our design motivation for counterfactual disambiguation in UTD. Second, the bottom row reveals an important limitation of subjective intention recognition. When the annotated label itself carries only 0.33 soft agreement, even the wrong prediction can remain compatible with the scene semantics. In this regime, SADIR should not be interpreted as making an arbitrary mistake; rather, it selects a nearby and still defensible intent hypothesis under intrinsically weak human consensus.

5. Discussion

5.1. A Stage-Aware View of Ambiguity

The experimental results support the central premise of this work: ambiguity in visual intention recognition is not a single phenomenon, but a stage-dependent one. During training, the main difficulty arises from supervisory ambiguity, where multiple plausible interpretations are compressed into unstable targets. During inference, the dominant challenge shifts to semantic ambiguity, where visually similar images must be separated among semantically adjacent intent labels. This explains why a single mechanism is often insufficient. UTD mainly improves the quality of representation learning under subjective supervision,

whereas SLR-C mainly improves local decision refinement among competing candidates.

This viewpoint also clarifies why a strong CLIP baseline already performs well. Once visual representation is sufficiently expressive, the remaining bottleneck is no longer basic visual recognition, but the mapping from rich visual semantics to abstract intent labels. In other words, the main challenge is not merely "seeing better", but "deciding better under ambiguity".

5.2. Heterogeneous Priors Improve Robustness Only with Controlled Integration

Our results show that heterogeneous priors are useful, but not in an unconstrained way. In SLR-C, the lexical, canonical, and scenario sources provide complementary signals: lexical phrases are concise but coarse, canonical descriptions are semantically cleaner but still abstract, and scenario descriptions offer the strongest visual grounding. Their combination improves semantic coverage and hard-case robustness. However, the component ablation also shows that a fixed heterogeneous prior alone is insufficient and may even distort the score geometry when directly injected without trainable correction.

This observation is important for understanding the role of SLR-C in the final system. The prior should not be interpreted as a standalone classifier. Instead, it should be viewed as a structured candidate refiner that proposes a more semantically informed local score space, after which the residual student learns to correct the remaining mismatch. This is precisely why the full system outperforms naive prior fusion, and why heterogeneous semantics become most effective when coupled with trainable adaptation.

5.3. Text Teacher Distillation Provides Structured Semantic Supervision

The rationale-related ablations reveal another important point: not every form of text supervision is equally useful. Generic image captions already provide some privileged signal, but they remain weaker than structured rationale. Likewise, simply enriching the text with contextual interpretation is not enough; the largest gain appears only when counterfactual disambiguation is included.

This suggests that the most useful textual signal for subjective intent recognition is not raw description, but a semantically sharpened explanation that explicitly contracts the boundary between nearby intents.

The modality ablation further reinforces this interpretation. A multimodal teacher with both image and text inputs is not the strongest distillation target, even though its oracle performance is not necessarily worse. This implies that the teacher is most useful when it behaves as a cleaner semantic supervisor rather than as another fused classifier. In this sense, UTD works less like conventional multimodal fusion and more like structured semantic regularization.

5.4. Limitations and Future Directions

Despite the consistent gains, the current framework still has several limitations. First, the method depends on offline text generation and external prior construction, which increases preparation cost even though UTD itself introduces no extra inference-time cost. Second, the gains shrink on weaker backbones such as ResNet101, suggesting that the current residual correction strategy still relies on reasonably strong base representations. Third, while the three-source prior improves robustness, its behavior also shows that richer priors are not automatically better unless their interaction with the student is carefully controlled.

These limitations suggest several future directions. One is to learn source-aware weighting instead of uniform prior averaging, so that the model can adaptively choose which semantic source to trust for each class or sample. Another is to design stronger student architectures for weaker backbones, making the stage-aware framework less dependent on CLIP-level representations. A third is to incorporate more explicit candidate-level reasoning, so that textual guidance operates not only through global distillation and local reranking, but also through confusion-pair-specific evidence resolution.

6. Conclusion

In this work, we studied subjective visual intention recognition from a stage-aware perspective and argued that supervisory ambiguity and semantic ambiguity should be handled differently across training and inference. Based on this view, we proposed SADIR, which combines Uncertainty-guided Text Distillation (UTD) for robust learning under annotator disagreement with Semantic Local Reranking and Calibration (SLR-C) for local semantic refinement at test time.

Extensive experiments on Intentonomy showed that the proposed framework consistently improves over strong visual baselines and prior state-of-the-art methods. Overall, the results suggest that once visual representations are sufficiently strong, the key challenge of intention understanding lies in resolving subjective and semantic ambiguity rather than in extracting low-level visual features alone. We hope this stage-aware formulation provides a clearer perspective for future research on human-centered visual understanding tasks.

Appendix A. Prompt Templates

For reproducibility, we include the exact prompt templates used to generate rationale supervision and scenario priors. These prompts are applied offline during data preparation and do not introduce additional inference-time cost.

Appendix A.1. Rationale Generation Prompt

We use an instruction template to generate the three-stage rationale for each training image. The final prompt used in our experiments is omitted here for brevity and will be inserted in the camera-ready version. The current placeholder structure is shown below.

In this template, y_{true} denotes the ground-truth intent set of the current sample, and $y_{confuse}$ denotes a semantically neighboring intent used for counterfactual disambiguation.

> **System Instruction:** You are an expert human behavioral analyst and computer vision specialist. Your task is to analyze an image and

provide a highly logical, structured reasoning chain that explains the underlying human intent. Strictly follow the requested format and avoid conversational filler.

User Prompt: Given this image, the ground-truth human intents are strictly defined as the concurrent occurrence of: y_{true}.

Please carefully observe the image and generate a structured reasoning report strictly following these 3 steps:

Step 1: Visual Evidence
Describe the explicit physical actions, key objects, body language, or facial expressions in the image that strongly support the presence of these concurrent intents (y_{true}). Be specific about what you see and how the visual cues correspond to these multiple intents.

Step 2: Contextual Bridging
Explain how the background, environment, or the relationship between the subjects logically connects with the visual cues from Step 1 to reveal why these multiple psychological motivations (y_{true}) naturally co-exist in this specific scene.

Step 3: Counterfactual Disambiguation
In a purely visual context, a machine learning model might easily misclassify this image as also containing the intent of $y_{confuse}$. Point out the specific visual clues that are definitively missing, or the contradictory details that are present, which prove the intent of $y_{confuse}$ is not happening here.

Appendix A.2. Scenario Prompt for SLR-C Prior

We also use an instruction template to generate scenario descriptions for each intent category, which are later encoded as textual priors in SLR-C.

For each intent class, we run this template once to obtain a concise scenario summary and a structured attribute list; the resulting text is then encoded by the frozen text encoder and used in local reranking.

Role You are a top-tier Computer Vision (CV) expert, a Prompt Engineer specializing in multimodal models (e.g., CLIP, BLIP), and an expert in visual psychology.

Task Your task is to leverage your extensive world knowledge to generate highly rich, concrete, and visually discriminative descriptions (Visual Anchors) for a specific intent category. These descriptions will serve as text queries to help a Vision-Language model accurately classify images.

Target Intent The current intent category to describe is: "Being good looking, attractive."

Output Guidelines To fully cover the implicit visual elements, please construct your description strictly across the following 5 dimensions. Translate the abstract intent into concrete visual anchors:

Visual Subjects & Objects: List 3 to 5 typical objects or subject combinations that strongly imply this intent (e.g., for "exploration," do not just say a backpack; include a topographical map, an off-road vehicle, a DSLR camera, or mud-splattered boots).

Actions & Interactions: Describe specific body language, micro-expressions, or ongoing physical interactions using strong action verbs (e.g., leaning in to observe closely, raising hands in triumph, staring intently at a screen).

Settings & Context: To address "intra-class variance," provide at least 3 distinct typical scenarios (e.g., indoor vs. outdoor, daily life vs. extreme environments) where this intent naturally occurs.

Emotion, Vibe & Atmosphere: What is the overall atmosphere conveyed by the image? What are the specific facial expressions or mood indicators? (e.g., a serene gaze, a warm and cozy domestic vibe, high-energy excitement).

Photography Style & Lighting: What visual style typically charac-

terizes web images of this intent? (e.g., high-contrast natural sunlight, warm indoor ambient lighting, vibrant and saturated colors, candid documentary style).

Format Requirement First, output a 50-80 word [Comprehensive Summary]. This summary should seamlessly integrate the visual elements above into a natural, cohesive, and highly visual paragraph. This will serve as my core Query. Second, provide a concise list breaking down the keywords/phrases for the 5 dimensions mentioned above.

Crucial Note: While generating, actively consider the core visual differences between this intent and other highly similar intents (e.g., distinguishing "Enjoying life" from "Having an easy and comfortable life"), and heavily emphasize those unique visual discriminators in your description.

Appendix A.2.1. Caption Prompt for Rational Ablation

For the rationale ablation in Section 4, we additionally use the following lightweight caption prompt as a weaker semantic supervision source.

System Instruction: You are an expert human behavioral analyst and computer vision specialist. Your task is to analyze an image and provide a highly logical, structured reasoning chain that explains the underlying human intent. Strictly follow the requested format and avoid conversational filler.

User Prompt: Describe this image for visual intent recognition in 2-3 concise sentences. Focus on visible actions, salient objects, and scene context. Avoid speculation beyond what is visually grounded.

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